



SYMBIOSIS INSTITUTE OF TECHNOLOGY, PUNE
Constituent of Symbiosis International (Deemed University), Pune

Name of Student: Anurag Thakur

PRN:25070122040

Semester: III

Year : AY 2026-27

Subject: Programming with Java

ASSIGNMENT No : 11

TITLE : Interface Implementation

Problem Statement

Write a Java program to demonstrate the implementation of interfaces.

Learning Objectives

To implement interfaces for achieving abstraction and multiple inheritance of type.

Theory

An interface specifies a set of abstract methods that implementing classes must define. It provides complete abstraction and supports multiple inheritance of type. Interfaces improve modularity, loose coupling, and software flexibility. They allow different classes to implement common behavior while maintaining independent implementations. Interfaces are widely used in enterprise application development.

Programme

1. Create an interface **Printable** and implement it in Student and Employee classes.

```
interface Printable {  
  
    void printDetails();  
}  
  
// Student class  
class Student implements Printable {
```

```
String name = "Rakesh";
int rollNo = 101;

public void printDetails() {
    System.out.println("Student Name: " + name);
    System.out.println("Roll Number: " + rollNo);
}
}

// Employee class
class Employee implements Printable {

    String name = "Amit";
    int employeeId = 501;

    public void printDetails() {
        System.out.println("Employee Name: " + name);
        System.out.println("Employee ID: " + employeeId);
    }
}

public class PrintableApplication {

    public static void main(String[] args) {

        Student s = new Student();

        System.out.println("Student Details:");
        s.printDetails();

        System.out.println();

        Employee e = new Employee();
```

```
        System.out.println("Employee Details:");
        e.printDetails();
    }
}
```

2. Create an interface **Switchable** with a method `turnOn()`. Implement the interface in **Light** and **Fan** classes to display the device status.

```
interface Switchable {

    void turnOn();
}

// Light class
class Light implements Switchable {

    public void turnOn() {
        System.out.println("Light is ON");
    }
}

// Fan class
class Fan implements Switchable {

    public void turnOn() {
        System.out.println("Fan is ON");
    }
}

public class SwitchableApplication {

    public static void main(String[] args) {

        Light light = new Light();

        System.out.println("Light Status:");
```

```

        light.turnOn();

        System.out.println();

        Fan fan = new Fan();

        System.out.println("Fan Status:");
        fan.turnOn();
    }
}

```

Output

```

cd "/Users/rakeshkumarchaudhary/assignment11/" && javac S
● rakeshkumarchaudhary@RAKESHs-MacBook-Air assignment11 % c
● rakeshkumarchaudhary@RAKESHs-MacBook-Air assignment11 % c
PrintableApplication
Student Details:
Student Name: Rakesh
Roll Number: 101

Employee Details:
Employee Name: Amit
Employee ID: 501
○ rakeshkumarchaudhary@RAKESHs-MacBook-Air assignment11 %

● rakeshkumarchaudhary@RAKESHs-MacBook-Air assignment11 % c
cd "/Users/rakeshkumarchaudhary/assignment11/" && javac S
● rakeshkumarchaudhary@RAKESHs-MacBook-Air assignment11 % c
a SwitchableApplication
Light Status:
Light is ON

Fan Status:
Fan is ON
○ rakeshkumarchaudhary@RAKESHs-MacBook-Air assignment11 %

```

Conclusion

In this practical, I learned how to use **interfaces in Java** to achieve abstraction and multiple inheritance of type. I implemented the Printable interface in Student and Employee classes and the Switchable interface in Light and Fan classes. This helped me understand how interfaces provide a common structure and allow different classes to implement the same methods in their own way.